iregi-rust benchmark

Batched one-loop tensor reduction in Rust versus IREGI and MadLoop

2026-10-23 · Timing environment not yet recorded

Reading guide. Every timing ratio is *iregi-rust time ÷ reference time*. Green means faster, white parity and red slower; timings are per phase-space point. Gray cells are unmeasured and amber cells are pending verification. Source inspection does not establish numerical agreement. The recorded ggH triangle path, current twelve-check CLI smoke suite and all31 frozen oracle cases pass; one oracle case requires explicit128/256-bit rescue. Higher-point accuracy, physical reference disagreements and mixed D0 derivatives remain open. No qualified timing is claimed.

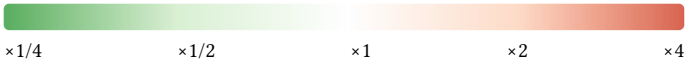
Acceptance gates

Correctness	Agreement with original IREGI and isolated MadLoop graphs	**Pending**
Pure reduction	Equal or better performance on identical inputs	**Unmeasured**
Complete evaluation	Report preparation, numerator, reduction and masters separately	**Unmeasured**

1 Identical-input comparisons

Table 1: **Synthetic inputs versus original IREGI.** Each measured comparison must share routing, masses, scale, rank, numerator coefficients and reduction mode. The current cells contain no measurements.

Case	n / r	Reduction	Masters	Total
Bubble Tensor components and CLI smoke pass	2 / 0–7	**Unmeasured** vs IREGI	**Unmeasured** vs OneLOop	**Unmeasured** vs IREGI
Triangle Tensor components and CLI smoke pass	3 / 0–7	**Unmeasured** vs IREGI	**Unmeasured** vs OneLOop	**Unmeasured** vs IREGI
Box Tensor components; full flow pending	4 / 0–7	**Unmeasured** vs IREGI	**Unmeasured** vs OneLOop	**Unmeasured** vs IREGI



2 Physical diagram comparisons

Integrated top triangle verified. Graph-isolated top-loop $gg \to H$ records are available for helicities 1 and 4, with 35 numerator coefficients and 105 tensor Laurent entries per graph. All four selected FeynKit numerators agree: top triangle, box and pentagon, plus a massless up-quark box, at the recorded real/complex probes. R2 and UV counterterms are excluded; bare UV/IR poles and R1 remain. The complete $gg \to H$ path agrees: 60 tensor slots have maximum normalized error 1.86×10^{-15}; the contracted finite coefficient differs by 9.38×10^{-15} in the recorded executable. Current CLI smoke checks pass; other integrated graphs and timings remain pending.

3 Validation coverage

Table 2: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass.	**Verified**
Scalar-master component	OneLoopMaster A0/B0/C0/D0/DB0 10 current tests pass, including lightlike B0/DB0 jets; generated native kernels pass specified scalar parity checks and remain opt-in.	**Verified**
IBP component	Powers, dimension shifts and singular identities 21 exact tests; independent Gamma oracle passes 200 tensor requests and 40 generalized requests, all four slots.	**Verified**
PV component	Every tensor moment and native Laurent slot 8 exact tests; ranks0–7 n1–4 pass. N5 rank6: Rust/IREGI f64 errors2.23e−10/3.40e−10 vs converged high precision; gate failed.	**Partial**
Precision and Dual	binary64, DoubleFloat, Float, static/dynamic jets Float128/256/512, specified second-order jets and grouping-identity regression pass; full physical mixed D0 derivatives remain pending.	**Partial**
Execution / CLI / core oracle	Batches through reduction and scalar masters 14 execution +8 contraction +6 CLI units +12 current smoke checks pass. Oracle31/31:30 binary64 +1 explicit128/256-bit rescue.	**Partial**
Exceptional kinematics	Explicit errors and verified fallback Stability10, exact bubble Taylor4 and precision-rescue3 pass. Full N5 rank6 rescue timed out at180s and remains unqualified.	**Partial**
Graph/numerator preparation	FeynKit DOT to Symbolica coefficients 4 exact model-boundary,13 preparation and4 physical numerator tests pass. Model-import regression fixed; fresh12-check CLI smoke passes.	**Verified**
Physical MadLoop comparison	Graph-isolated top loops and massless box Four numerators and integrated ggH pass. Other integrated graphs pending; original ggHHH modes disagree by0.60013 despite stable flags.	**Partial**

Snapshot boundary. Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Master precision checks include binary64, DoubleFloat and Float at 128, 256 and 512 bits. Verified jets include static/dynamic second-order tadpoles and Euclidean massive triangle/box scaling derivatives. General physical first, second and mixed derivatives remain under audit. **Open precision gate:** for the five-propagator rank-six case, 128- and 256-bit evaluation converge. Binary64 Rust and original IBP differ from that value by 2.23×10^{-10} and 3.40×10^{-10} respectively. Both exceed the fixed 10^{-10} criterion and need validated precision rescue. The frozen oracle passes all 31 cases: 30 at binary64 and one generalized triangle through explicit 128/256-bit rescue. The five-point rank-six rescue attempt timed out after 180 seconds and remains unqualified.

3.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator q_x^4, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

Method	Finite coefficient / i
Parameter integral	45.4316300015058951832997…
Original IBP	45.43163000150602
Original PV / hybrid	46.11123944517427

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant Rust comparisons use original IBP and the independent parameter-integral result.

4 Measurement protocol

Run release builds on one pinned CPU core with fixed thread counts and record hardware, operating system, toolchain, compile flags, dependency revisions, input hashes, seeds, warm-up, sample count and raw observations. Every result must identify whether caches are cold or warm. Report median and dispersion over repeated sufficiently long timed blocks; retain an output checksum.

Preparation comprises graph import, rules, tensor identities, coefficient extraction and evaluator construction. Time numerator evaluation, pure reduction, scalar masters and complete evaluation independently. Measure the complete total directly: adding separately instrumented components is only a diagnostic. Distinguish reference-library calls from subprocess and file I/O.

Use the same samples at batch widths 1, 4, 8, 16, 32 and 64 and report time per point. For precision comparisons, match the numerical precision and include rescue costs. Record failed lanes; do not omit unfavorable points silently. Absolute/scaled coefficient errors, high-precision adjudication and derivative checks accompany performance results. No speed gate passes without correctness. The stable binary64 target is a mixed absolute/relative tolerance of 10^{-10} on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput separately for each topology/rank/mass family. Record IBP, PaVe and hybrid mode independently. The eventual reference envelope is seven propagators and rank seven, including generalized powers and dimension shifts.

5 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `typst compile docs/benchmark.typ docs/benchmark.pdf`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

5.1 Validation reproduction and remaining checks

The standalone checks reported here run with the pinned dependencies prepared by `scripts/env.sh`. The master suite uses `RUST_MIN_STACK=67108864 target/iregi-masters-tests --test-threads=1 --nocapture` the IBP suite uses `target/ibp-unit --test-threads=1 --nocapture`. The current checkpoint passes ten master, twenty-one IBP, eight PV exact, fourteen execution, eight contraction, ten stability and six CLI unit tests. Twelve current CLI smoke checks pass, including model import and the reference comparison command. The frozen oracle passes all 31 cases: 30 at binary64 and `C0_power2_1_1_rank2` through explicit batched 128/256-bit precision rescue with unchanged tolerance. This is not 31 binary64 successes. Three rescue tests cover exact promotion, generalized triangle coefficients, second-order Dual values and invalid-lane isolation. The full five-point rank-six rescue test timed out after 180 seconds; its numerical qualification remains open. Four exact bubble-Taylor tests also pass; see `crates/iregi-masters/src/bubble_neighborhood.rs`.

Generated instruction-major Rust master kernels remain opt-in through `native-batch`. Scalar parity passes for all five families with binary64 batches of 0, 1, 3, 5 and 31 lanes, including an invalid lane; DoubleFloat/Float parity passes for A0–D0. A grouping regression preserves zero-primal jets and genuine conditional branches. These checks do not qualify physical mixed D0 derivatives or establish runtime speed.

An independent Dirichlet-simplex Gamma formula, expanded by Symbolica, matches 200 tensor-coefficient scalar requests through rank seven and 40 generalized power/dimension requests in all four native Laurent slots. Run `cargo test -p iregi-rust --test triangle_gamma -- --test-threads=1`. This is exact algebra for the massless triangle with one off-shell leg; it is not a broad numerical or performance result.

PV numerical checks compare all tensor moments and native pole slots using original scalar-master leaf values. The strict, dimensionally normalized comparison passes all 330 moments through rank seven for one to four propagators, including both complex components. The first open failure is $N = 5$, rank six, numerator $q_0^5 q_3$. Both binary64 codes exceed the unchanged criterion against the converged 128/256-bit result; the numerical values and rescue timeout remain visible in the open qualification ledger. Evidence and unqualified higher-point cases remain in `validation/pv-high-rank/rust-coverage.json`. A genuinely singular six-propagator point is an explicit error, not a successful reduction. The scalar IBP identities include selected singular triangles, coincident denominators and lightlike bubbles. A standalone six-dimensional pentagon needs information beyond the finite orders of the A0–D0 basis and must produce an explicit unsupported-domain diagnostic when that information is unavailable.

6 Physical fixture qualification

Single graph	Pointwise numerator	Integrated Rust result
Top triangle: $gg \to H$	**Verified**	**Verified**
Top box: $gg \to HH$	**Verified**	**Pending**
Top pentagon: $gg \to HHH$	**Verified**	**Pending**
Massless up-quark box	**Verified**	**Pending**

6.1 Physical reference contract

The export selects two oriented top triangles with loop grouping disabled. Fixtures preserve model parameters, actual HELAS polarizations, graph factors, source hashes and numerical probes. Original MadLoop uses denominator shifts $(0, p_1, p_1 + p_2)$ with a polynomial in $x = -Q$; its convention adapter already includes the odd-rank sign. With the supplied FeynKit routing, numerator comparison is $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ after matching couplings and polarizations. Do not apply the fixed-color or direction factor twice.

The committed exports are `validation/physical/madloop/fixtures/ggh-helicity-1.json` and `ggh-helicity-4.json`. The compact numerator reference is `validation/physical/ggh-numerator-reference.json`. All thirteen preparation tests and all four selected physical numerator comparisons pass. The five schema-2 capture files contain 40 graph records, including the second triangle helicity. This is four selected-graph numerator checks, not 40 integrated validations. The pentagon discrepancy was resolved by exporting MadLoop's actual evaluated external momenta; requested driver momenta can differ after phase-space correction. The corrected massless-up binding also passes. Captured numerical results were unchanged by the metadata correction.

The complete $gg \to H$ CLI path passes at the recorded binary64 point. All 15 tensor moments / 60 Laurent slots agree with original IREGI at the FeynKit canonical routing to 1.86×10^{-15} normalized error. The fixed-color raw finite coefficient is approximately -147.86843349622455; its MadLoop comparison error is 9.38×10^{-15}, with matching single and double poles. The normalized finite result is approximately -0.9363877940734862. Values, small imaginary residuals, hashes and reproduction commands are retained in `validation/physical/ggh-cli-comparison.json`. The other three integrated graph comparisons remain pending. These references are binary64; higher working precision alone cannot certify additional digits. Short, unpinned timing smoke runs are excluded from benchmark evidence. The recorded ggH result retains its exact tested executable hash. The current CLI smoke suite passes twelve checks; the physical result is not a fresh replay. A model-import regression was resolved by a narrow Symbolica-backed quantum-number adapter. Four boundary, thirteen preparation and four physical numerator tests pass with the fix; other numerical tokens retain their precision.

7 Open qualification ledger

Unresolved item	Observed evidence and remaining check
Five-point rank-six accuracy	Rust binary64 error 2.23×10^{-10}; original IBP error 3.40×10^{-10} versus the converged 128/256-bit value. Both fail the unchanged 10^{-10} gate.
Full pentagon rescue	Three active rescue tests pass, including the generalized triangle. The full rank-six pentagon regression timed out after 180 seconds; its success is unqualified.
Physical pentagon reference	Original IBP and PV/hybrid disagree by 0.6001338887 normalized error at rank four, although every mode reports stability. The correct result has not been adjudicated.
Physical mixed D0 derivatives	The full mixed-derivative suite remains pending. Euclidean scaling and scalar batch parity are narrower checks.
Runtime performance	No qualified kernel timing, speedup or parity result has been measured. Short unpinned timing smoke runs remain excluded.

7.1 Canonical physical reference audit

At the FeynKit canonical input for $gg \to HHH$, original IBP and PV/hybrid disagree in a finite rank-four tensor coefficient. The largest normalized difference is 0.6001338886545732 at moment index 54; all modes claim stability. This reference-only audit does not identify which result is correct and does not run Rust reduction. The other three physical families agree across original modes within 2.22×10^{-14}. Inputs, hashes and all four Laurent-slot comparisons are retained in `validation/physical/canonical-reference-audit.json` reproduce with `validation/physical/audit_reference.py`.

The retained `validation/pv-high-rank/reference-coverage.json` also records rank-seven original PV instability for five to seven propagators and a falsely stable four-propagator disagreement. These reference failures remain visible; stable IBP reference values do not themselves certify Rust high-rank coverage.

7.2 Independent integral evidence

The rank-four bubble check is reproducible with `python3 validation/pv-high-rank/parameter_check.py --write`. Its 50- and 80-digit quadratures agree with original IBP. The suspected quadratic-root parenthesization error is a source diagnosis; the numerical discrepancy is established independently of that diagnosis. Independent lightlike-bubble quadratures at 60 and 100 digits, including first invariant derivatives, are retained in `validation/pv-high-rank/lightlike-feynman-parameters.json`. The five-propagator precision probe is retained in `validation/pv-high-rank/precision-probe.json` its source and reproduction command are recorded with the complete input and 128/256-bit results. Correctly rounded master inputs reduce the observed finite-component error to 3.11×10^{-12} with 256-bit reduction and 2.23×10^{-11} with binary64 reduction. This isolates amplified scalar-master error; it does not qualify the original failing binary64 route.

8 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

8.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j = \left(q+p_j\right)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

8.2 Native poles and MadLoop conversion

The native result order is

$$(U,I,J,F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master (L_0, L_1, L_2), the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U,I,J,F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

8.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

9 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

9.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

9.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

9.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Primary references: Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.